[SCHOOL SEAL] [DIVISION SEAL] [DepEd SEAL]

Republic of the Philippines
Department of Education
Region V – Division of [PROVINCE]
[SCHOOL NAME IN BOLD CAPS]
[Municipality], [Province]

KALINGA: A LOW-COST BAYMAX-INSPIRED WEARABLE HEALTH PENDANT WITH VITAL-SIGN CHECKS, FALL ALERTS, TIMED PILL DISPENSING, AND A FAMILY MONITORING APP

Submitted to the Scientific Review Committee as an Official Entry to the
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Science Innovation Expo (SIE) – Team Category

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PATENT / COPYRIGHT APPLICATION

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The IPOPHL application or certificate (Utility Model, Industrial Design, or Copyright) must already have been applied for and secured before the entry is accepted at the Regional Fair (SIE entries only).

ISEF FORMS

[INSERT HERE]

International Rules for Pre-College Science Research forms, signed by the research adviser and the researchers. The forms that apply to this project are listed below.

Form	Title	Applies?
1	Checklist for Adult Sponsor	Yes
1A	Student Checklist	Yes
—	Research Plan / Project Summary	Yes
1B	Approval Form	Yes
3	Risk Assessment Form	Yes
2	Qualified Scientist Form	Only if done in a research institution
4	Human Participants Form	Yes (IRB approval and informed consent)
5A / 5B	Vertebrate Animal Forms	No
6A / 6B	Potentially Hazardous Biological Agents	No
7	Continuation / Research Progression	No

RESEARCH PLAN

A. Rationale

Many older adults and people with chronic illness stay home alone while their families work or live abroad.

Falls are a leading cause of injury and death among adults over 60, and long waits for help make them worse (World Health Organization [WHO], 2021).

About half of patients on long-term medicines do not take them as prescribed (WHO, 2003).

Smartwatches, medical alert services, and pill dispensers each cover only part of the problem, and together they cost more than most families can pay.

Low-cost sensors, 3D printing, and free cloud databases now make it possible to put these jobs into one wearable that the family can watch from a phone.

B. Research Question or Problem

This study aimed to develop and evaluate KALINGA, a low-cost Baymax-inspired wearable health pendant with vital-sign checks, fall alerts, timed pill dispensing, and a family monitoring app.

1. How closely do KALINGA's temperature, pulse rate, and SpO_2 readings agree with reference instruments (mean bias, 95% limits of agreement, mean absolute error, and Arms)?
2. How accurately does KALINGA detect simulated falls and ignore daily activities (sensitivity and specificity), and how fast do fall, SOS, and inactivity alerts reach the app?
3. How accurately and punctually does the pill carousel present each scheduled dose, and how well does it record doses as taken, late, or missed?
4. What share of readings and alerts reach the app, and how long does one battery charge last?
5. How do older adults and family caregivers rate the usability of the pendant and the app?
6. How much does KALINGA cost compared with commercial products that do the same jobs?

C. Goals, Expected Outcomes, and Hypotheses

The project is expected to produce a working pendant and app and to show whether the pendant's readings, alerts, and doses are accurate and timely enough for home monitoring at a lower cost.

- $H_0$1: There is no significant difference between KALINGA's readings and the reference readings for temperature, pulse rate, and SpO_2.
- $H_0$2: KALINGA's fall decision is not associated with the true event type (fall or daily activity).
- $H_0$3: The mean System Usability Scale score of KALINGA is not significantly different from 68, the average score of products in general.

D. Procedures

1. Design the pendant shell in CAD and 3D-print it in PLA.
2. Wire the ESP32-C3, sensors, servo, OLED, buzzer, vibration motor, and battery.
3. Program the pendant; set up the Firebase database and the KALINGA Monitor app.
4. Calibrate the thermometer, pulse oximeter, and fall thresholds on a separate calibration set.
5. Run agreement trials with consenting volunteers against reference instruments.
6. Simulate falls (researchers only, on gym mats) and daily activities; time the alerts.
7. Run five days of scheduled doses, battery tests, and app delivery checks.
8. Run usability sessions with older adults and family caregivers.

E. Risk and Safety

- Human participants: IRB approval and informed consent before testing; non-invasive checks only.
- Simulated falls only by the researchers onto 20-cm gym mats, with pads and adult supervision.
- LiPo battery: protected charger, no puncture or heat; charging only under supervision.
- 3D printer and soldering: adult supervision, ventilation, heat-resistant gloves.
- Health data: private Firebase account, coded ids, deleted after the fair (RA 10173).
- Pills in testing: placebo candy tablets only; no real medicines are dispensed.

F. Data Analysis

- Bland–Altman bias and 95% limits of agreement, MAE, Arms; paired t-test ($H_0$1).
- Confusion matrix, sensitivity, specificity; chi-square test of independence ($H_0$2).
- Percentages for correct slot, on-time doses, delivered messages; mean battery life.
- System Usability Scale; one-sample t-test against 68 ($H_0$3). $\alpha = 0.05$; Excel and JASP.

G. Bibliography

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Photographs, flowcharts, and schematics carry a bold left-aligned label instead of a numbered caption and are not listed here.

ABSTRACT

Many older adults and people with chronic illness spend the day alone while their families work, and a fall, a missed dose, or a rising fever may go unnoticed for hours. Smartwatches, medical alert services, and pill dispensers each solve one part of the problem and together cost more than most families can pay. Inspired by the soft healthcare companion of the animated film Big Hero 6, this study developed KALINGA, a 3D-printed wearable pendant built around an ESP32-C3 microcontroller. It checks forehead temperature with a non-contact infrared sensor and pulse rate and SpO_2 with a fingertip pad, detects falls and long inactivity with an accelerometer, has an SOS button, and turns a four-slot pill carousel to present each scheduled dose. Readings, doses, and alerts are sent through Wi-Fi to a Firebase database and shown live in the KALINGA Monitor app on the family's phones. Against reference instruments, the pendant's readings had mean biases of ____ °C, ____ bpm, and ____% SpO_2 (SpO_2 Arms = ____%). It detected ____% of simulated falls with ____% specificity for daily activities, and alerts reached the app in ____ s. The carousel presented the correct slot in ____% of ____ doses, ____% on time. The battery lasted ____ h. Users gave a System Usability Scale score of ____ (p = ____ against 68). The pendant cost PHP ____. KALINGA brings health checks, medicine timing, and emergency alerts into one affordable wearable that families can watch from anywhere.

Keywords: wearable health monitor; fall detection; pill dispenser; pulse oximetry; Internet of Things; elderly care

I. INTRODUCTION

Many older adults and people with chronic illness stay home alone while their families work long hours or abroad (Republic Act No. 9994, 2010).

Falls are a leading cause of injury and death among adults over 60, and lying on the floor for a long time makes the harm worse (WHO, 2021).

About half of patients on long-term medicines miss or mistime their doses (WHO, 2003).

Smartwatches check pulse and detect falls, medical alert pendants call for help, and automatic dispensers time pills, but each does only one part and many need monthly fees.

[NEED: the specific family, barangay, or senior citizens' group where the researchers observed the problem, what was observed, and when. Community Connection and Impact is scored in the Science Innovation Expo.]

There is a need for one low-cost wearable that checks basic vital signs, times medicines, and alerts the family from anywhere.

In response to the aforementioned problems, the researchers aimed to develop KALINGA, a 3D-printed, Baymax-inspired health pendant with vital-sign checks, fall and SOS alerts, a four-slot pill carousel, and the KALINGA Monitor app, and to evaluate its accuracy, timeliness, usability, and cost.

Features and Specification

- Product: KALINGA, a white, rounded, Baymax-inspired pendant worn on a breakaway neck cord, with an original two-dot face design.
- Size and weight: about 60 × 83 × 39 mm; ____ g with battery and pills.
- Vital-sign checks: forehead temperature (MLX90614, non-contact), pulse rate and SpO_2 (MAX30102 finger pad); twice a day on schedule, on demand by the OK button, or when the family taps Check now.
- Safety: fall detection and no-movement alert (MPU6050), SOS button, 10-second cancel window for false alarms.
- Medicine: four-slot pill carousel turned by an SG90 micro servo; each slot opens at its scheduled time with vibration, beeps, and a message on the 0.91-in OLED; the user confirms with OK.

- App: KALINGA Monitor, an installable web app showing live status, alerts, vital-sign trends, the medicine schedule, and the dose log; works on Android, iPhone, and computers.
- Power: 3.7 V 1000 mAh LiPo battery charged by USB-C; target of one day per charge.
- Intended use: home monitoring of older adults and people with chronic illness by their families. It screens and alerts; it does not diagnose or treat.

Market Trends and Opportunities

The number of Filipino families with an older member at home grows each year, and more family members work away from home. Smartwatches with SpO_2 and fall detection, medical alert pendants with monitoring services, and automatic pill dispensers are sold online, but a family needs several of them to cover what KALINGA does. Table 1 compares their costs. Prices are to be filled in from the researchers' own canvass of local stores.

Table 1

Cost Comparative Analysis of KALINGA and Commercial Products

Product	Price (PHP)	Vital checks	Fall/SOS	Timed dose	Family app
KALINGA pendant + app	2,150 (est.)	Yes	Yes	Yes	Yes, free
Smartwatch with SpO_2 and fall detection	____	Yes	Some	No	Yes
Medical alert pendant with service	____ + ____/month	No	Yes	No	Some
Automatic pill dispenser	____	No	No	Yes	Some
Pulse oximeter and thermometer	____	Yes	No	No	No

Note. Table created by the researchers, 2026.

Scientific Basis

Pulse oximeters shine red and infrared light through a fingertip; the ratio of light absorbed with each heartbeat gives SpO_2, and the beat interval gives pulse rate (Allen, 2007; Jubran, 2015). ISO 80601-2-61 accepts an SpO_2 Arms of 4% or less (International Organization for Standardization [ISO], 2017b).

Every warm surface gives off infrared energy that rises with its temperature (Fraden, 2016). The MLX90614 measures this energy from the forehead; because skin is cooler than the

body core, a calibration equation is applied, as in clinical forehead thermometers tested under ISO 80601-2-56 (ISO, 2017a).

A fall shows as a short drop in acceleration (free fall), a sharp peak (impact), and then stillness. Thresholds on these three phases detect falls with high sensitivity and specificity (Bourke et al., 2007).

Timed reminders and pre-sorted doses are among the simplest ways to improve medicine adherence (WHO, 2003). A new measuring device must be compared with a reference using limits of agreement, not correlation alone (Bland & Altman, 1986), and usability can be measured with the 10-item System Usability Scale, whose average across products is 68 (Brooke, 1996; Sauro, 2011).

Statement of the Problem

This study aimed to develop and evaluate KALINGA. Specifically, it sought to answer the following questions:

1. How closely do KALINGA's temperature, pulse rate, and SpO_2 readings agree with reference instruments (mean bias, 95% limits of agreement, mean absolute error, and Arms)?
2. How accurately does KALINGA detect simulated falls and ignore daily activities (sensitivity and specificity), and how fast do fall, SOS, and inactivity alerts reach the app?
3. How accurately and punctually does the pill carousel present each scheduled dose, and how well does it record doses as taken, late, or missed?
4. What share of readings and alerts reach the app, and how long does one battery charge last?
5. How do older adults and family caregivers rate the usability of the pendant and the app?
6. How much does KALINGA cost compared with commercial products that do the same jobs?

Hypotheses

The following null hypotheses were tested at the 0.05 level of significance:

- $H_0 1$: There is no significant difference between KALINGA's readings and the reference readings for temperature, pulse rate, and SpO_2.

- $H_0$2: KALINGA's fall decision is not associated with the true event type (fall or daily activity).
- $H_0$3: The mean System Usability Scale score of KALINGA is not significantly different from 68, the average score of products in general.

II. MATERIALS AND METHODS

Procurement of the Materials

Components were bought from local electronics shops and online stores; the shell was printed on a ____ 3D printer [NEED: printer model, or the print shop used]. The reference instruments were borrowed from the school clinic. Expenses are in Appendix A.3.

Table 2

Materials and Components of KALINGA

Material / Component	Quantity	Purpose
ESP32-C3 SuperMini	1	Controller, Wi-Fi
MLX90614 infrared thermometer	1	Forehead temperature
MAX30102 pulse oximeter module	1	Pulse rate and SpO_2
MPU6050 accelerometer	1	Falls and inactivity
0.91-in OLED display	1	Time, doses, results
SG90 micro servo and printed carousel	1	Pill dispensing
Vibration motor and buzzer	1 each	Reminders and alarms
LiPo 1000 mAh, TP4056 USB-C charger	1 each	Power
White PLA, black paint	60 g	3D-printed shell
Reference thermometer and pulse oximeter	1 each	Reference readings (borrowed)
20-cm gym mats, knee and elbow pads	2; 1 set	Simulated falls

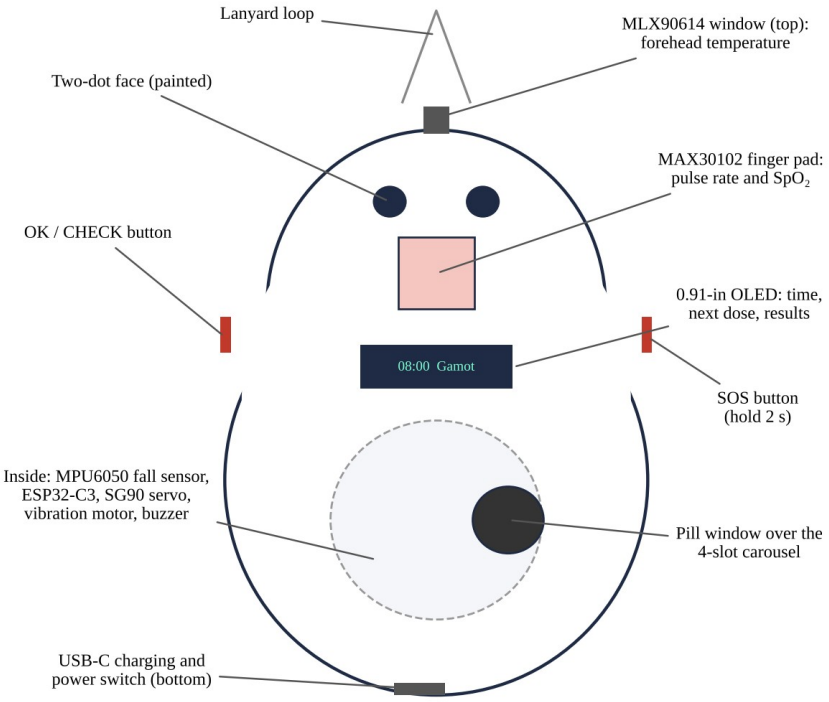
Note. Table created by the researchers, 2026.

Construction of the Product

Design and 3D Printing of the Shell

The shell was modeled as two parts, a front shell and a back shell, plus a pill carousel, with a parametric script that outputs STL files (Appendix, Dimensions of the Product). The parts were printed in white PLA with a 0.4-mm nozzle, 0.2-mm layers, 3 walls, and 20% infill, face-down on the bed with tree supports. The eyes were painted with black acrylic in the two recesses, and the edges were sanded.

Design of the KALINGA Pendant



Front view, actual size about 60 × 83 × 39 mm, 3D-printed in white PLA.

Note. Image created by the researchers, 2026.

3D-Printed Parts: Back Shell, Pill Carousel, and Front Shell

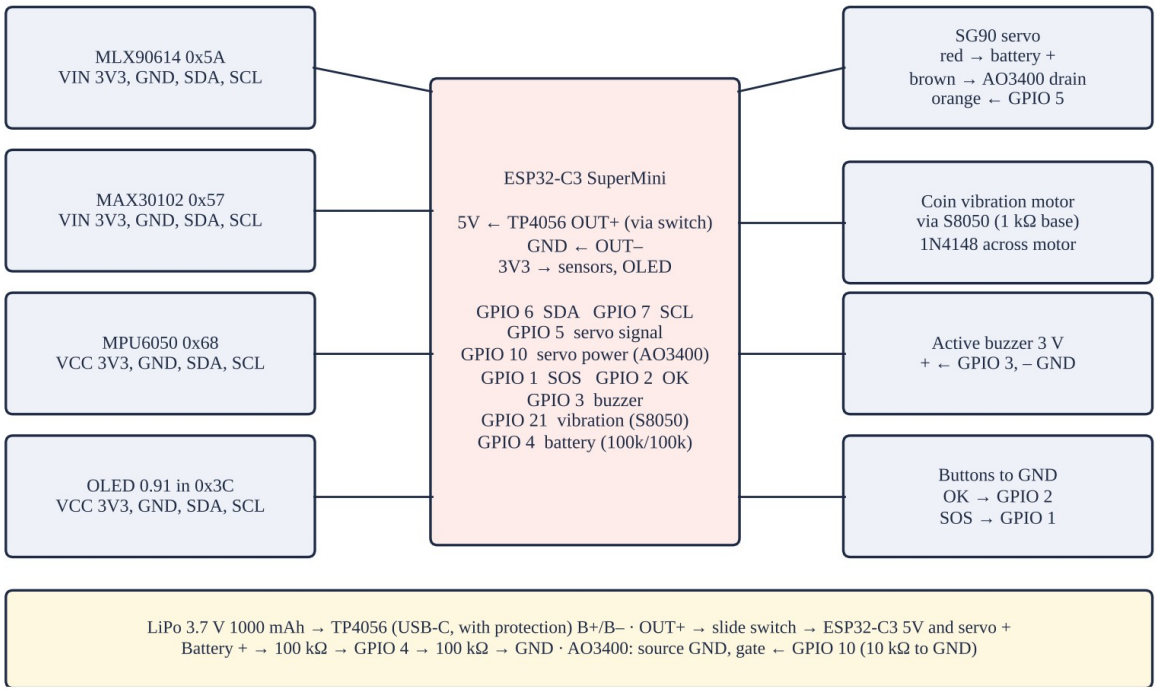


Note. Image created by the researchers, 2026.

Electronics

The MLX90614, MAX30102, MPU6050, and OLED share one I²C bus at 100 kHz on the ESP32-C3. The SG90 servo is powered from the battery through an AO3400 MOSFET that cuts its power between doses. The vibration motor is driven by an S8050 transistor, and a 100 kΩ/100 kΩ divider lets the ESP32-C3 read the battery voltage. Connections are in Appendix A.4.

Wiring Diagram of KALINGA



Note. Image created by the researchers, 2026.

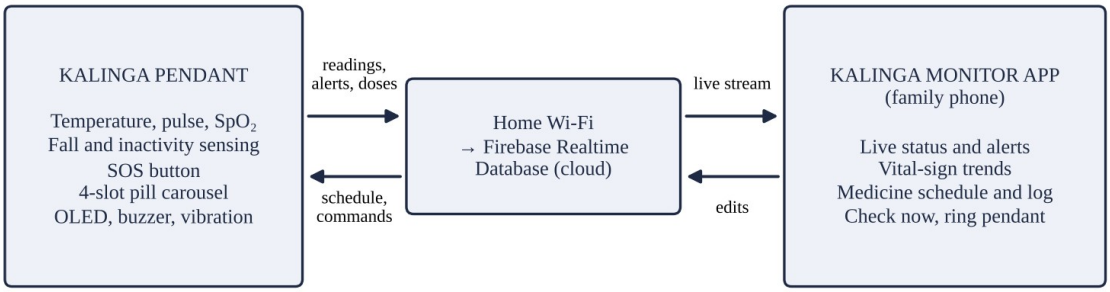
Pill Carousel

The 36-mm carousel has four pockets at 45°, 90°, 135°, and 180° and a solid sector at 0°, which closes the window. At each dose time the servo turns the right pocket under the window; the pendant vibrates and beeps every 2 minutes for up to 30 minutes until the user tips the pill out and presses OK. Doses are recorded as taken (within 10 minutes), late, or missed, and a missed dose alerts the family.

Program and App

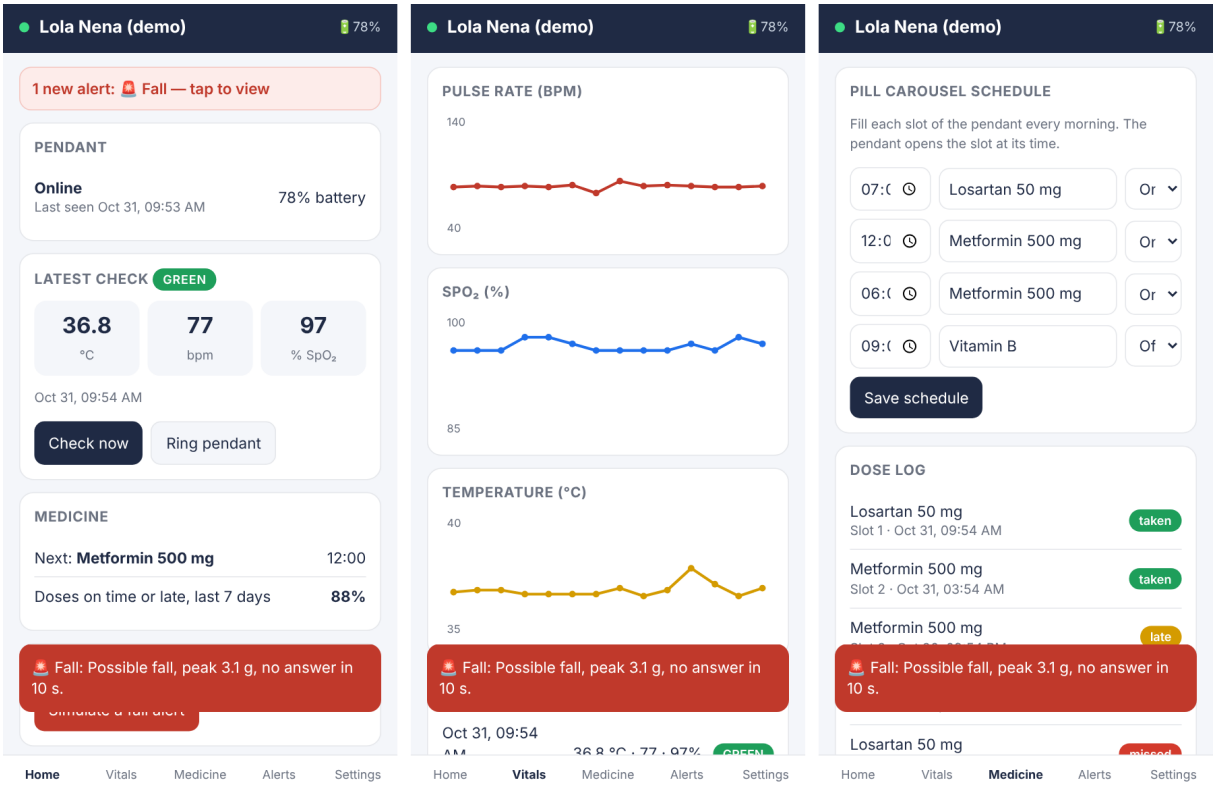
The ESP32-C3 checks for falls 50 times a second and, once a minute, sends its status and reads the schedule and commands from a Firebase Realtime Database through its REST interface (Google, n.d.). The KALINGA Monitor app signs in with Firebase Authentication and receives changes as a live stream, so new readings and alerts appear within seconds. Database rules let only the family's accounts read or write their patient's data.

How Data Flows Between the Pendant and the App



Note. Image created by the researchers, 2026.

KALINGA Monitor App (demo mode): Home, Vitals, and Medicine Screens



Note. Image created by the researchers, 2026.

Data Gathering

Tests were done indoors at 26 to 30 °C. Volunteers rested seated for 5 minutes before each check, and the pendant and the reference instrument were used within 1 minute of each other in alternating order. The first 10 volunteers formed the calibration set and were not used in

the agreement analysis. Simulated falls (forward, backward, sideways, and from a chair) were done only by the researchers onto 20-cm gym mats; daily activities were sitting, lying on a bed, bending, walking, and climbing a step. Placebo candy tablets were used in all dose tests.

Table 3

Trials Conducted in the Study

Trial set	What is measured	Number	Reference
Bench temperature	IR reading of a 35.0–40.0 °C target	6 × 10	Probe thermometer
Calibration set	Temperature, PR, SpO_2	10 volunteers × 3	Clinic instruments
Validation set	Temperature, PR, SpO_2	____ volunteers × 3	Clinic instruments
Falls and activities	Fall decision, alert time	____ falls; ____ activities	Video
Doses	Slot, timing, record	5 days × 4 slots	Observer log
Delivery and battery	Messages received; hours per charge	5 days; 3 charges	App log
Usability	SUS	____ older adults; ____ caregivers	Questionnaire

Note. Table created by the researchers, 2026.

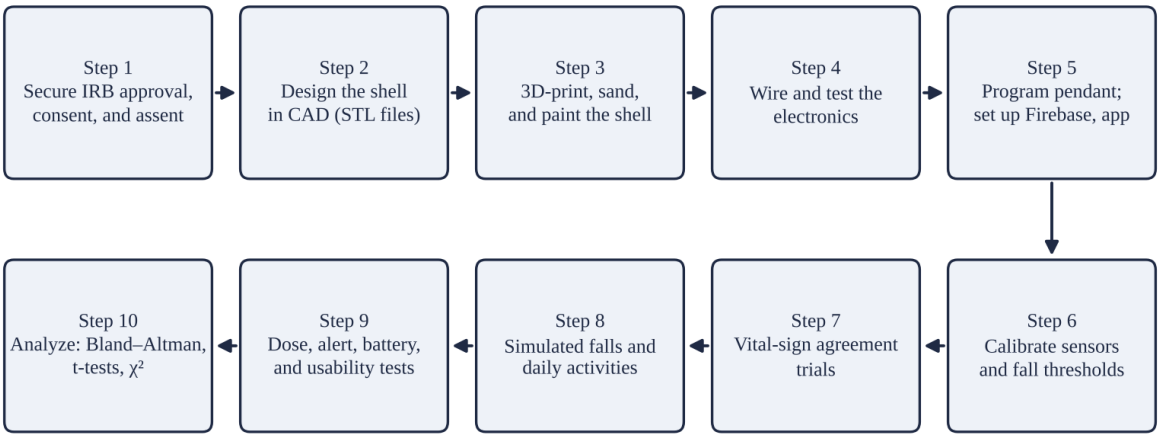
The following formulas were used:

$d = KALINGA - reference$; $Bias = \Sigma d \div n$	where difference and mean bias	(1)
$LoA = Bias \pm 1.96 \times SD(d)$; $Arms = \sqrt{(\Sigma d^2 \div n)}$	where limits of agreement; SpO_2 RMS error	(2)
$Sensitivity = TP \div (TP + FN)$; $Specificity = TN \div (TN + FP)$	where falls detected; activities ignored	(3)
$t_alert = t_app - t_event$	where alert time (s)	(4)
Correct-slot rate = correct slots ÷ doses × 100	where carousel accuracy (%)	(5)
$SUS = 2.5 \times [\Sigma(\text{odd item} - 1) + \Sigma(5 - \text{even item})]$	where System Usability Scale score (0–100)	(6)

$H_0$1 was tested with a paired t-test for each measure, $H_0$2 with a chi-square test of independence, and $H_0$3 with a one-sample t-test against 68, at $\alpha = 0.05$, using Microsoft Excel and JASP.

Flowchart of the Procedure

Steps of the Study



Note. Image created by the researchers, 2026.

III. RESULTS AND DISCUSSION

[NEED: every ____ in this chapter must be replaced with values the researchers measured. No value may be entered that was not measured.]

Agreement of the Vital-Sign Checks

Table 4.1 shows the infrared thermometer against a water-bath target, and Table 4.2 the agreement with the reference instruments in ____ validation volunteers. Temperature had a bias of ____ °C (LoA ____ to ____), pulse rate ____ bpm, and SpO_2 ____% with an Arms of ____%, which ____ the 4% limit. The paired t-test gave p = ____, ____, and ____, so $H_0 1$ is ____.

Table 4.1

Bench Test of the Infrared Thermometer Against a Water-Bath Target

Target (°C)	Probe reading (°C)	ALAGA mean ± SD (°C)	Error (°C)
35.0	____	____ ± ____	____
36.0	____	____ ± ____	____
37.0	____	____ ± ____	____
37.5	____	____ ± ____	____
38.0	____	____ ± ____	____
40.0	____	____ ± ____	____

Note. Table created by the researchers, 2026.

Table 4.2

Agreement of KALINGA With the Reference Instruments

Measure	n	Bias	95% LoA	MAE	Arms	p
Temperature (°C)	____	____	____ to ____	____	____	____
Pulse rate (bpm)	____	____	____ to ____	____	____	____
SpO_2 (%)	____	____	____ to ____	____	____	____

Note. Table created by the researchers, 2026.

Figure 1

Bland–Altman Plots of Temperature, Pulse Rate, and SpO_2

[NEED: Bland–Altman plots from the measured validation data: difference (ALAGA − reference) against the mean of the two, with lines for the bias and the 95% limits of agreement.]

Note. Figure created by the researchers, 2026.

Fall Detection and Alerts

KALINGA detected ____ of ____ simulated falls (sensitivity ____%) and ignored ____ of ____ daily activities (specificity ____%); χ^2 = ____, p = ____, so $H_0$2 is ____ (Table 5.1). Alerts reached the app in a mean of ____ s (Table 5.2).

Table 5.1

Confusion Matrix of KALINGA Fall Decisions

	True fall	Daily activity	Total
KALINGA: fall	TP = ____	FP = ____	____
KALINGA: no fall	FN = ____	TN = ____	____
Total	____	____	____

Note. Table created by the researchers, 2026.

Table 5.2

Time for Alerts to Reach the App

Alert	n	Mean ± SD (s)	Longest (s)
Fall	____	____ ± ____	____
SOS	____	____ ± ____	____
No movement	____	____ ± ____	____
Missed dose	____	____ ± ____	____
Abnormal check	____	____ ± ____	____

Note. Table created by the researchers, 2026.

Pill Carousel and Dose Records

Over ____ doses, the carousel presented the correct slot ____% of the time, ____% within 1 minute of schedule, and the dose log matched the observers' record in ____% of doses (Table 6.1).

Table 6.1

Dose Presentation and Records Over Five Days

Measure	Count	Result (%)
Correct slot presented	____ / ____	____
Presented within 1 min of schedule	____ / ____	____
Records matching the observer (taken, late, missed)	____ / ____	____
Missed-dose alerts received	____ / ____	____

Note. Table created by the researchers, 2026.

Data Delivery and Battery Life

____% of readings and ____% of alerts reached the app, and one charge lasted ____ h (Table 7.1).

Table 7.1

Data Delivery and Battery Life

Measure	Result
Readings received by the app	____ / ____ (____%)
Alerts received by the app	____ / ____ (____%)
Battery life per charge, mean ± SD (h)	____ ± ____
Charging time (h)	____

Note. Table created by the researchers, 2026.

Usability

Older adults gave a mean SUS score of ____ and caregivers ____; against 68, t = ____, p = ____, so $H_0$3 is ____ (Table 8.1).

Table 8.1

System Usability Scale Scores

Group	n	Mean ± SD	t (vs 68)	p
Older adults	____	____ ± ____	____	____
Family caregivers	____	____ ± ____	____	____
All users	____	____ ± ____	____	____

Note. Table created by the researchers, 2026.

Discussion

[NEED: explain the results against the literature: why the SpO_2 and temperature biases are as found (finger movement, skin tone, forehead sweat), which falls were missed and why, what caused late or missed doses, and what users liked or found hard. Compare cost with Table 1.]

IV. CONCLUSION

Summary of Findings

1. Readings had biases of ____ °C, ____ bpm, and ____% SpO_2, with an SpO_2 Arms of ____%; $H_0$1 is ____.
2. Falls were detected with ____% sensitivity and ____% specificity; alerts arrived in ____ s; $H_0$2 is ____.
3. The carousel presented the correct slot in ____% of doses, ____% on time.
4. ____% of readings and alerts reached the app; the battery lasted ____ h.
5. Users gave a SUS score of ____; $H_0$3 is ____.
6. KALINGA cost PHP ____, against PHP ____ for the commercial products in Table 1.

Conclusion

[NEED: one paragraph based on the measured results.] KALINGA joins vital-sign checks, fall and SOS alerts, timed doses, and family monitoring in one low-cost wearable.

Recommendations

1. Shrink the pendant with a custom circuit board and a smaller dispensing mechanism.
2. Add phone push notifications and an LTE module for use outside home Wi-Fi.
3. Test with real falls data and with people of different skin tones and ages.
4. File the shell as an Industrial Design and the carousel as a Utility Model with IPOPHL, keeping the design original rather than copying the film character.

V. ACKNOWLEDGEMENT

[NEED: thank the people and offices that helped: adviser, school clinic, volunteers, family, and any partner who printed the shell.]

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VII. APPENDICES

Appendix A. Raw Data, Expenses, and Connections

Table A.1

Raw Data Sheet for Vital-Sign Checks

Code	Trial	ALAGA T (°C)	Ref T (°C)	ALAGA PR	Ref PR	ALAGA SpO_2	Ref SpO_2	Pain	Time (s)
V01	1								
V01	2								
V01	3								
V02	1								
V02	2								
V02	3								

Note. Table created by the researchers, 2026.

Table A.2

Raw Data Sheet for Falls and Daily Activities

Trial	Event	Detected (Y/N)	Cancelled	Peak g	Alert time (s)	Notes
1	Forward fall					
2	Backward fall					
3	Sideways fall					
4	Fall from chair					
5	Sit down					
6	Lie on bed					
7	Bend					
8	Walk					
9	Climb a step					

Note. Table created by the researchers, 2026.

Table A.3

Expense Summary of the Study

Item	Qty	Estimated price (PHP)
ESP32-C3 SuperMini	1	220
MLX90614 (GY-906) module	1	550
MAX30102 module	1	180
MPU6050 (GY-521) module	1	120
0.91-in OLED, 128 × 32, I²C	1	150
SG90 micro servo	1	120
LiPo 3.7 V 1000 mAh (503450)	1	220
TP4056 USB-C charger with protection	1	45
Coin vibration motor, 3 V active buzzer	1 set	50
AO3400, S8050, 1N4148,	1 set	30

resistors		
Tactile buttons with caps, slide switch	1 set	35
Perfboard, wires, headers	1 lot	80
PLA filament or print service (about 60 g)	1 lot	250
Black acrylic paint, sandpaper	1 lot	40
Breakaway lanyard, M2 screws	1 set	60
Total (estimated)		**2,150**

Note. Table created by the researchers, 2026.

Table A.4

Pin Connections of KALINGA

Part	Pin	ESP32-C3	Note
MLX90614, MAX30102, MPU6050, OLED	SDA / SCL	GPIO 6 / 7	I²C at 100 kHz, 3.3 V
SG90 servo	signal / power	GPIO 5 / GPIO 10	AO3400 switches the servo ground
OK and SOS buttons	to GND	GPIO 2 / GPIO 1	Internal pull-ups
Buzzer; vibration motor	+ ; S8050 base	GPIO 3; GPIO 21	1 kΩ base resistor; 1N4148
Battery divider	100k/100k	GPIO 4	Reads 0–4.2 V
TP4056 OUT+ / OUT–	via slide switch	5V / GND	LiPo on B+ / B–

Note. Table created by the researchers, 2026.

Dimensions of the Product

Overall size about 60 mm wide, 83 mm tall, and 39 mm deep, including the lanyard loop; pill carousel 36 mm across with four 9.6-mm pockets. Print files: cad/front_shell.stl, cad/back_shell.stl, and cad/pill_carousel.stl, made by cad/kalinga_pendant.py.

Front Shell, Rendered From the STL File



Note. Image created by the researchers, 2026.

[NEED: photographs of the finished pendant with dimension callouts and its weight.]

Program and App

Pendant program: firmware/kalinga_pendant/kalinga_pendant.ino (ESP32C3 Dev Module). App: app/index.html with its manifest and service worker; database rules in app/database.rules.json. The engineering notebook gives every build and setup step.

Photographs of the Procedure

[NEED: dated photographs of each step in the flowchart, each with a bold label and the note “Photographs taken by the researchers, 2026.” No faces of volunteers without written permission.]

Informed Consent

Summary read to each participant: “You are invited to try KALINGA, a pendant that checks temperature, pulse, and oxygen and reminds you about medicine. You will sit while it reads your forehead from a short distance and you place one finger on its pad; the clinic's instruments will be used too. You will answer a short questionnaire. It takes about 20 minutes and does not hurt. You will not be asked to fall or take any medicine. Your name will not be recorded.”

[NEED: signed consent and assent forms and the IRB approval.]

Curriculum Vitae

[NEED: one-page curriculum vitae of each researcher with a 2 × 2 photo.]